Group 7 Project

G7

2025-11-25

Data Import and Description

```
library(readxl)
data <- read_excel("ENB2012_data.xlsx")
head(data)
```

```
## # A tibble: 6 × 10
##      X1    X2    X3    X4    X5    X6    X7    X8    Y1    Y2
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  0.98  514.  294   110.     7     2     0     0  15.6  21.3
## 2  0.98  514.  294   110.     7     3     0     0  15.6  21.3
## 3  0.98  514.  294   110.     7     4     0     0  15.6  21.3
## 4  0.98  514.  294   110.     7     5     0     0  15.6  21.3
## 5  0.9   564.  318.  122.     7     2     0     0  20.8  28.3
## 6  0.9   564.  318.  122.     7     3     0     0  21.5  25.4
```

```
library(dplyr)
# Rename all columns once at the top
colnames(data) <- c(
  "Relative_Compactness",
  "Surface_Area",
  "Wall_Area",
  "Roof_Area",
  "Overall_Height",
  "Orientation",
  "Glazing_Area",
  "Glazing_Distribution",
  "Heating_Load",
  "Cooling_Load")

# Drop Cooling_Load
data <- data[ , !(names(data) %in% "Cooling_Load") ]

glimpse(data)
```

```
## Rows: 768
## Columns: 9
## $ Relative_Compactness <dbl> 0.98, 0.98, 0.98, 0.98, 0.90, 0.90, 0.90, 0.90, 0…
## $ Surface_Area         <dbl> 514.5, 514.5, 514.5, 514.5, 563.5, 563.5, 563.5, …
## $ Wall_Area            <dbl> 294.0, 294.0, 294.0, 294.0, 318.5, 318.5, 318.5, …
## $ Roof_Area            <dbl> 110.25, 110.25, 110.25, 110.25, 122.50, 122.50, 1…
## $ Overall_Height       <dbl> 7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0,…
## $ Orientation          <dbl> 2, 3, 4, 5, 2, 3, 4, 5, 2, 3, 4, 5, 2, 3, 4, 5, 2…
## $ Glazing_Area         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…
## $ Glazing_Distribution <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…
## $ Heating_Load         <dbl> 15.55, 15.55, 15.55, 15.55, 20.84, 21.46, 20.71, …
```

```
head(data)
```

```
## # A tibble: 6 × 9
##   Relative_Compactness Surface_Area Wall_Area Roof_Area Overall_Height
##                  <dbl>        <dbl>     <dbl>     <dbl>          <dbl>
## 1                 0.98         514.       294      110.              7
## 2                 0.98         514.       294      110.              7
## 3                 0.98         514.       294      110.              7
## 4                 0.98         514.       294      110.              7
## 5                 0.9          564.       318.     122.              7
## 6                 0.9          564.       318.     122.              7
## # ℹ 4 more variables: Orientation <dbl>, Glazing_Area <dbl>,
## #   Glazing_Distribution <dbl>, Heating_Load <dbl>
```

```
summary(data)
```

```
##  Relative_Compactness  Surface_Area     Wall_Area       Roof_Area
##  Min.   :0.6200       Min.   :514.5   Min.   :245.0   Min.   :110.2
##  1st Qu.:0.6825       1st Qu.:606.4   1st Qu.:294.0   1st Qu.:140.9
##  Median :0.7500       Median :673.8   Median :318.5   Median :183.8
##  Mean   :0.7642       Mean   :671.7   Mean   :318.5   Mean   :176.6
##  3rd Qu.:0.8300       3rd Qu.:741.1   3rd Qu.:343.0   3rd Qu.:220.5
##  Max.   :0.9800       Max.   :808.5   Max.   :416.5   Max.   :220.5
##  Overall_Height  Orientation    Glazing_Area    Glazing_Distribution
##  Min.   :3.50   Min.   :2.00   Min.   :0.0000   Min.   :0.000
##  1st Qu.:3.50   1st Qu.:2.75   1st Qu.:0.1000   1st Qu.:1.750
##  Median :5.25   Median :3.50   Median :0.2500   Median :3.000
##  Mean   :5.25   Mean   :3.50   Mean   :0.2344   Mean   :2.812
##  3rd Qu.:7.00   3rd Qu.:4.25   3rd Qu.:0.4000   3rd Qu.:4.000
##  Max.   :7.00   Max.   :5.00   Max.   :0.4000   Max.   :5.000
##   Heating_Load
##  Min.   : 6.01
##  1st Qu.:12.99
##  Median :18.95
##  Mean   :22.31
##  3rd Qu.:31.67
##  Max.   :43.10
```

check for nulls in the dataset

```
colSums(is.na(data))
```

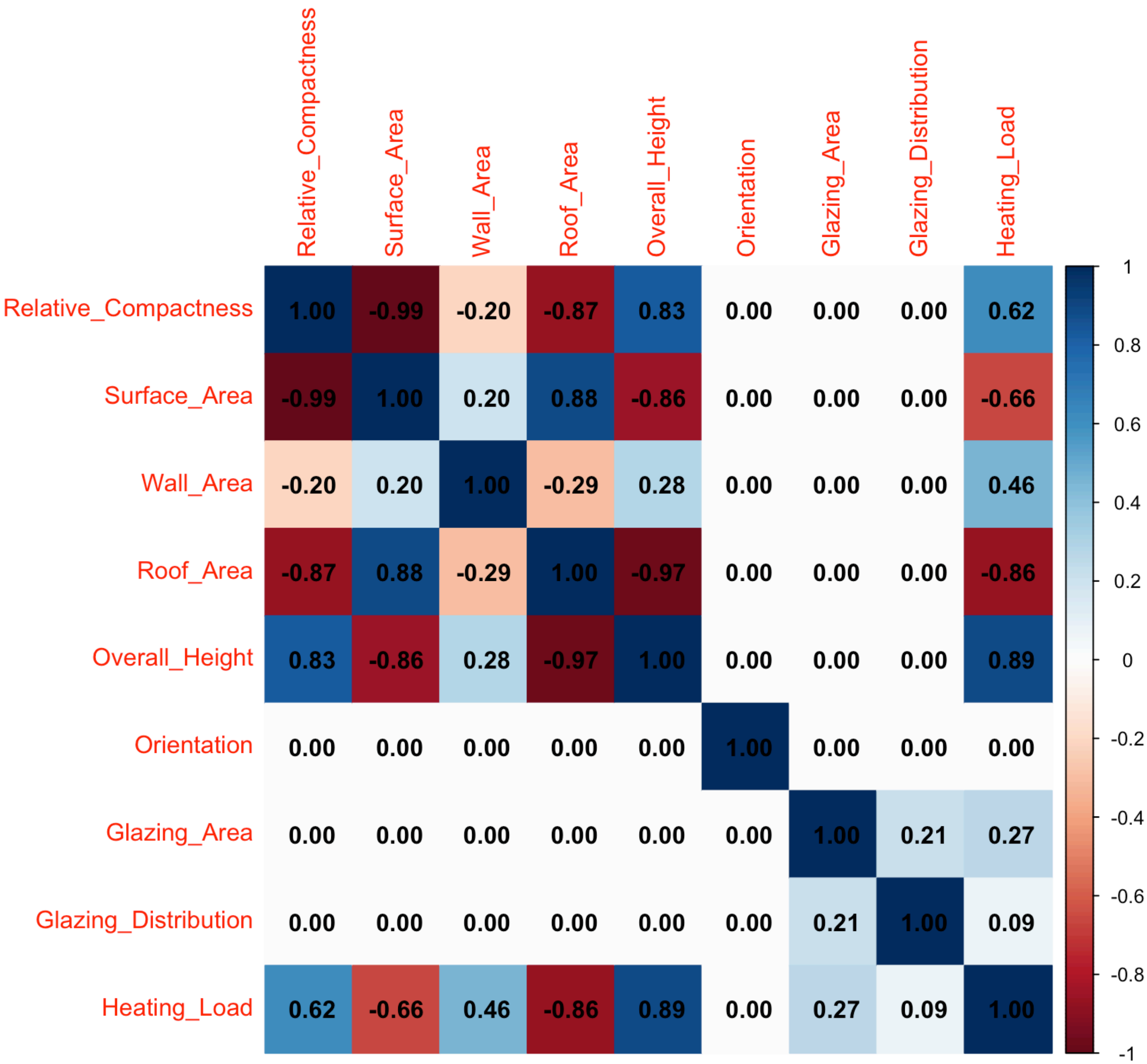
```
## Relative_Compactness         Surface_Area            Wall_Area
##                    0                    0                    0
##            Roof_Area       Overall_Height          Orientation
##                    0                    0                    0
##         Glazing_Area Glazing_Distribution         Heating_Load
##                    0                    0                    0
```

#correlation matrix

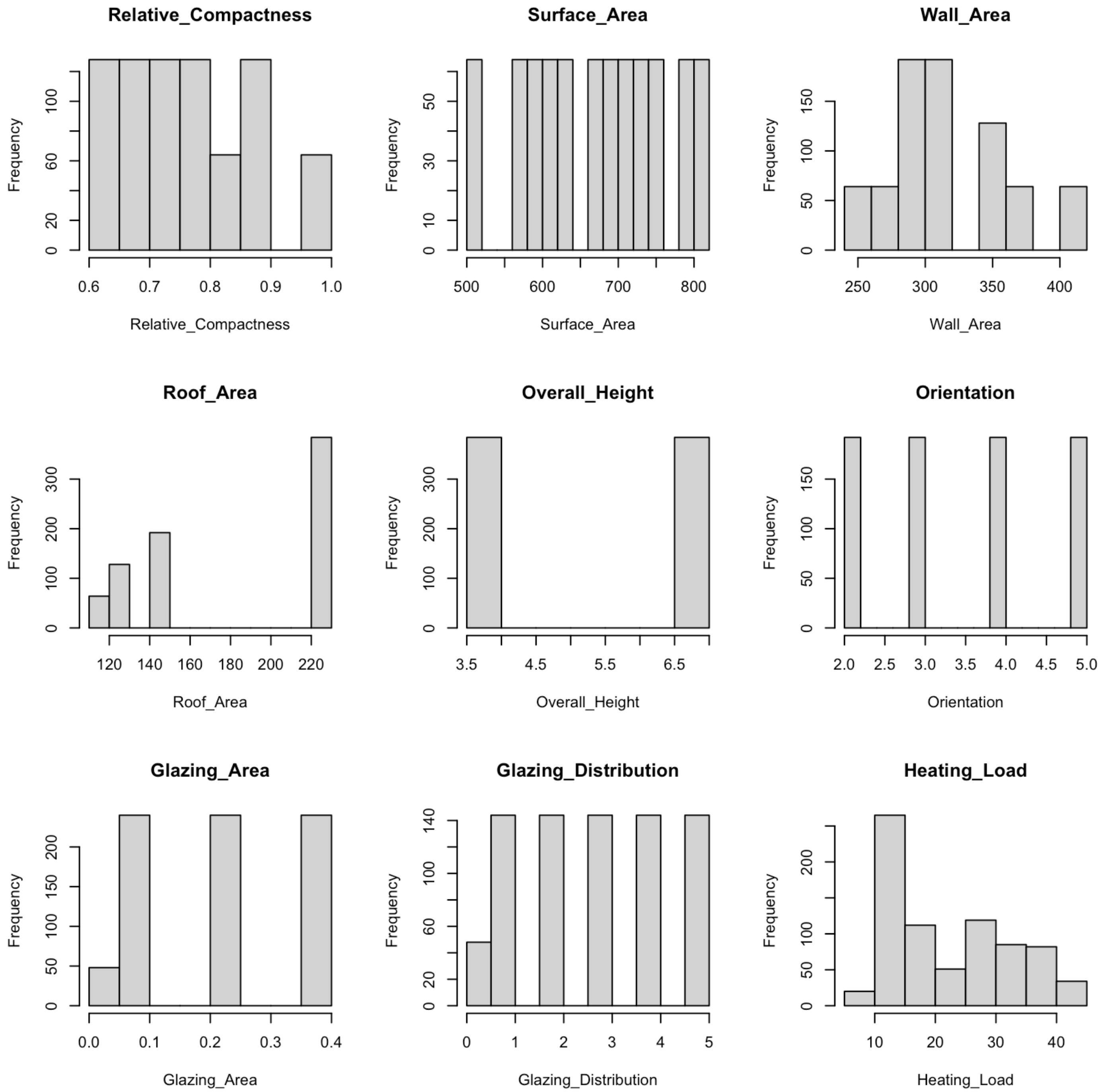
```
num_data <- data[sapply(data, is.numeric)]
cor_matrix <- cor(num_data)
cor_matrix
```

```
##                      Relative_Compactness  Surface_Area  Wall_Area
## Relative_Compactness         1.000000e+00 -9.919015e-01 -0.2037817
## Surface_Area                -9.919015e-01  1.000000e+00  0.1955016
## Wall_Area                   -2.037817e-01  1.955016e-01  1.0000000
## Roof_Area                   -8.688234e-01  8.807195e-01 -0.2923165
## Overall_Height               8.277473e-01 -8.581477e-01  0.2809757
## Orientation                  0.000000e+00  0.000000e+00  0.0000000
## Glazing_Area                 7.617400e-20  4.664140e-20  0.0000000
## Glazing_Distribution         0.000000e+00  0.000000e+00  0.0000000
## Heating_Load                 6.222719e-01 -6.581199e-01  0.4556714
##                          Roof_Area Overall_Height  Orientation  Glazing_Area
## Relative_Compactness -8.688234e-01      0.8277473  0.000000000  7.617400e-20
## Surface_Area          8.807195e-01     -0.8581477  0.000000000  4.664140e-20
## Wall_Area            -2.923165e-01      0.2809757  0.000000000  0.000000e+00
## Roof_Area             1.000000e+00     -0.9725122  0.000000000 -1.197187e-19
## Overall_Height       -9.725122e-01      1.0000000  0.000000000  0.000000e+00
## Orientation           0.000000e+00      0.0000000  1.000000000  0.000000e+00
## Glazing_Area         -1.197187e-19      0.0000000  0.000000000  1.000000e+00
## Glazing_Distribution  0.000000e+00      0.0000000  0.000000000  2.129642e-01
## Heating_Load         -8.618281e-01      0.8894305 -0.002586763  2.698417e-01
##                      Glazing_Distribution Heating_Load
## Relative_Compactness           0.00000000  0.622271936
## Surface_Area                   0.00000000 -0.658119917
## Wall_Area                      0.00000000  0.455671365
## Roof_Area                      0.00000000 -0.861828052
## Overall_Height                 0.00000000  0.889430464
## Orientation                    0.00000000 -0.002586763
## Glazing_Area                   0.21296422  0.269841685
## Glazing_Distribution           1.00000000  0.087368460
## Heating_Load                   0.08736846  1.000000000
```

```
# Plot correlation heatmap
corrplot::corrplot(cor_matrix,method = "color",addCoef.col = "black")
```

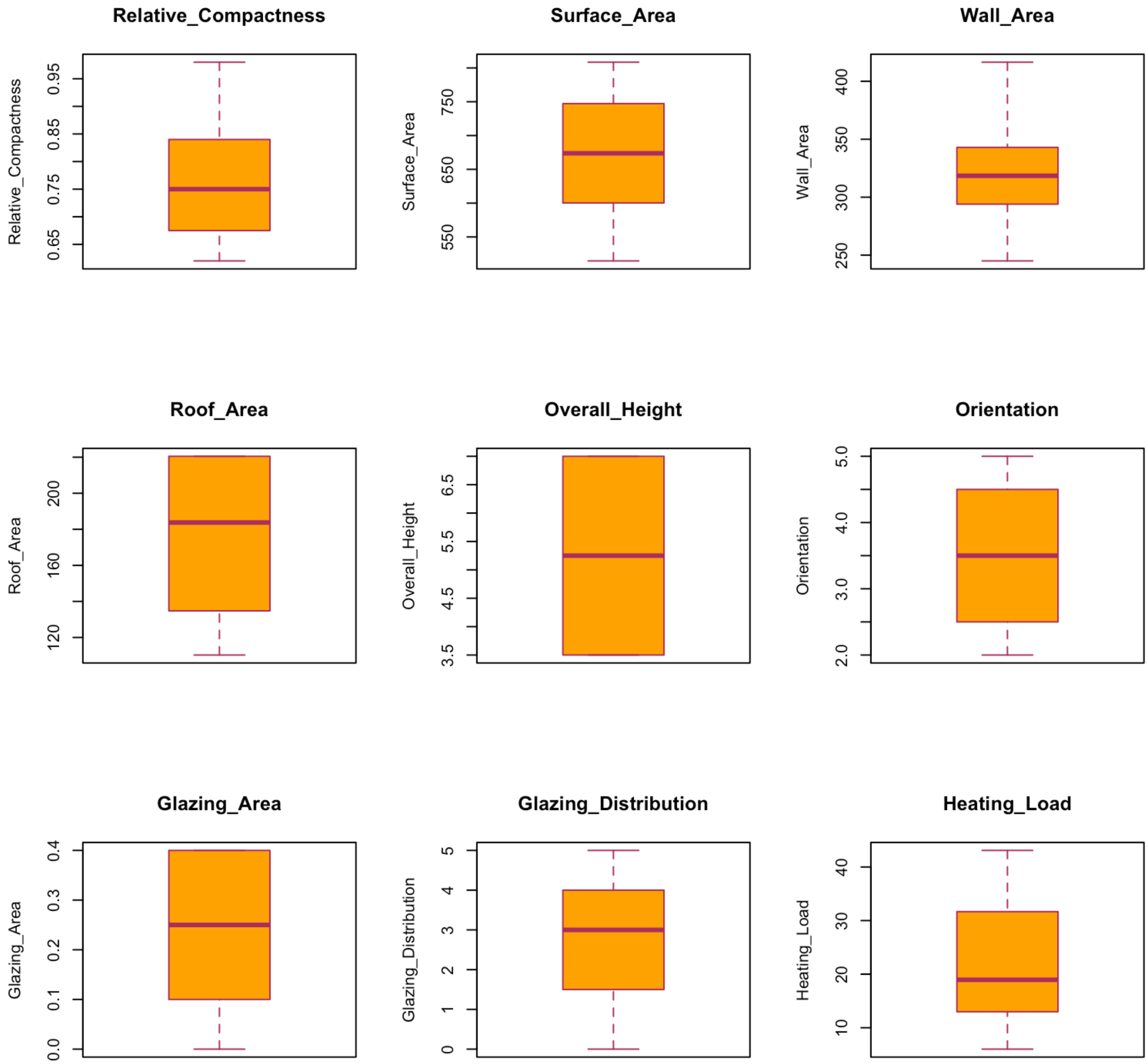



```
par(mfrow=c(3,3))
for (v in names(data)) {
  hist(data[[v]], main=v, xlab=v)
}
```




```
par(mfrow=c(1,1))
```

```
par(mfrow=c(3,3))
for (v in names(data)) {
  boxplot(data[[v]],
          main=v,
          ylab=v,
         col = "orange",
         border = "maroon")
}
```




```
par(mfrow=c(1,1))
```

```
model1 <- lm(Heating_Load ~ Relative_Compactness, data = data)
summary(model1)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.569  -6.332  -1.028   3.393  19.259
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -23.053      2.081  -11.08   <2e-16 ***
## Relative_Compactness   59.359      2.698   22.00   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.904 on 766 degrees of freedom
## Multiple R-squared:  0.3872, Adjusted R-squared:  0.3864
## F-statistic:   484 on 1 and 766 DF,  p-value: < 2.2e-16
```

```
AIC(model1)
```

```
## [1] 5358.922
```

```
model2 <- lm(Heating_Load ~ Relative_Compactness + Surface_Area, data = data)
summary(model2)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area,
##     data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.509  -5.184  -0.386   5.433  15.377
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           355.23626   30.37456  11.695   <2e-16 ***
## Relative_Compactness -180.46408   19.37575  -9.314   <2e-16 ***
## Surface_Area           -0.29034    0.02327 -12.479   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.209 on 765 degrees of freedom
## Multiple R-squared:  0.4909, Adjusted R-squared:  0.4895
## F-statistic: 368.8 on 2 and 765 DF,  p-value: < 2.2e-16
```

```
AIC(model2)
```

```
## [1] 5218.632
```

```
model3 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area,
             data = data)
summary(model3)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.0800  -2.0566  -0.1426   2.3888  12.0990
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           2.665e+02  1.794e+01   14.86   <2e-16 ***
## Relative_Compactness -1.459e+02  1.138e+01  -12.82   <2e-16 ***
## Surface_Area         -2.625e-01  1.364e-02  -19.24   <2e-16 ***
## Wall_Area             1.369e-01  3.574e-03   38.30   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.221 on 764 degrees of freedom
## Multiple R-squared:  0.8257, Adjusted R-squared:  0.825
## F-statistic:  1206 on 3 and 764 DF,  p-value: < 2.2e-16
```

```
AIC(model3)
```

```
## [1] 4397.545
```

```
model4 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area +
               Roof_Area,
             data = data)
summary(model4)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area + Roof_Area, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.0800  -2.0566  -0.1426   2.3888  12.0990
##
## Coefficients: (1 not defined because of singularities)
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           2.665e+02  1.794e+01   14.86   <2e-16 ***
## Relative_Compactness -1.459e+02  1.138e+01  -12.82   <2e-16 ***
## Surface_Area         -2.625e-01  1.364e-02  -19.24   <2e-16 ***
## Wall_Area             1.369e-01  3.574e-03   38.30   <2e-16 ***
## Roof_Area                    NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.221 on 764 degrees of freedom
## Multiple R-squared:  0.8257, Adjusted R-squared:  0.825
## F-statistic:  1206 on 3 and 764 DF,  p-value: < 2.2e-16
```

```
AIC(model4)
```

```
## [1] 4397.545
```

```
model5 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area +
               Roof_Area +
               Overall_Height,
             data = data)
summary(model5)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area + Roof_Area + Overall_Height, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1764  -2.2608   0.1508   2.4278  10.6491
##
## Coefficients: (1 not defined because of singularities)
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           89.176620  26.041444   3.424 0.000649 ***
## Relative_Compactness -64.773432  14.081121  -4.600 4.95e-06 ***
## Surface_Area          -0.087289   0.023368  -3.735 0.000201 ***
## Wall_Area              0.060813   0.009098   6.684 4.47e-11 ***
## Roof_Area                    NA         NA      NA       NA
## Overall_Height         4.169954   0.462540   9.015  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.016 on 763 degrees of freedom
## Multiple R-squared:  0.8424, Adjusted R-squared:  0.8416
## F-statistic:  1020 on 4 and 763 DF,  p-value: < 2.2e-16
```

```
AIC(model5)
```

```
## [1] 4321.807
```

```
model6 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area +
               Roof_Area +
               Overall_Height +
               Orientation,
             data = data)
summary(model6)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area + Roof_Area + Overall_Height + Orientation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1414  -2.2741   0.1576   2.4180  10.6374
##
## Coefficients: (1 not defined because of singularities)
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           89.258276  26.061926   3.425 0.000648 ***
## Relative_Compactness -64.773432  14.090058  -4.597 5.01e-06 ***
## Surface_Area          -0.087289   0.023383  -3.733 0.000203 ***
## Wall_Area              0.060813   0.009103   6.680 4.60e-11 ***
## Roof_Area                    NA         NA      NA       NA
## Overall_Height         4.169954   0.462833   9.010  < 2e-16 ***
## Orientation           -0.023330   0.129686  -0.180 0.857280
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.018 on 762 degrees of freedom
## Multiple R-squared:  0.8425, Adjusted R-squared:  0.8414
## F-statistic: 814.9 on 5 and 762 DF,  p-value: < 2.2e-16
```

```
AIC(model6)
```

```
## [1] 4323.774
```

```
model7 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area +
               Roof_Area +
               Overall_Height +
               Orientation +
               Glazing_Area,
             data = data)
summary(model7)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area + Roof_Area + Overall_Height + Orientation + Glazing_Area,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3512  -1.3812  -0.0121   1.3291   7.5438
##
## Coefficients: (1 not defined because of singularities)
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           84.468127  19.126463   4.416 1.15e-05 ***
## Relative_Compactness -64.773432  10.339990  -6.264 6.26e-10 ***
## Surface_Area          -0.087289   0.017159  -5.087 4.59e-07 ***
## Wall_Area              0.060813   0.006681   9.103  < 2e-16 ***
## Roof_Area                    NA         NA      NA       NA
## Overall_Height         4.169954   0.339650  12.277  < 2e-16 ***
## Orientation           -0.023330   0.095170  -0.245    0.806
## Glazing_Area          20.437968   0.799220  25.572  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.949 on 761 degrees of freedom
## Multiple R-squared:  0.9153, Adjusted R-squared:  0.9146
## F-statistic:  1370 on 6 and 761 DF,  p-value: < 2.2e-16
```

```
AIC(model7)
```

```
## [1] 3849.45
```

```
model8 <- lm(Heating_Load ~
               Relative_Compactness +
               Surface_Area +
               Wall_Area +
               Roof_Area +
               Overall_Height +
               Orientation +
               Glazing_Area +
               Glazing_Distribution,
             data = data)
summary(model8)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Relative_Compactness + Surface_Area +
##     Wall_Area + Roof_Area + Overall_Height + Orientation + Glazing_Area +
##     Glazing_Distribution, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.8965 -1.3196 -0.0252  1.3532  7.7052
##
## Coefficients: (1 not defined because of singularities)
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          84.013418  19.033613   4.414 1.16e-05 ***
## Relative_Compactness -64.773432  10.289448  -6.295 5.19e-10 ***
## Surface_Area          -0.087289   0.017075  -5.112 4.04e-07 ***
## Wall_Area              0.060813   0.006648   9.148  < 2e-16 ***
## Roof_Area                    NA         NA      NA       NA
## Overall_Height         4.169954   0.337990  12.338  < 2e-16 ***
## Orientation           -0.023330   0.094705  -0.246  0.80548
## Glazing_Area          19.932736   0.813986  24.488  < 2e-16 ***
## Glazing_Distribution   0.203777   0.069918   2.915  0.00367 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.934 on 760 degrees of freedom
## Multiple R-squared:  0.9162, Adjusted R-squared:  0.9154
## F-statistic:  1187 on 7 and 760 DF,  p-value: < 2.2e-16
```

```
AIC(model8)
```

```
## [1] 3842.914
```

AICs (forward selection)

```
AIC(model1, model2, model3, model4, model5, model6, model7, model8)
```

```
##        df      AIC
## model1  3 5358.922
## model2  4 5218.632
## model3  5 4397.545
## model4  5 4397.545
## model5  6 4321.807
## model6  7 4323.774
## model7  8 3849.450
## model8  9 3842.914
```

Note that model 4 had no difference with model 3, Roof_Area is collinear to surfce area

Forward Regression Model

```
#intercept only
base_model <- lm(Heating_Load ~ 1, data = data)

#full scope
full_scope <- formula(lm(Heating_Load ~ ., data = data))


reg_model <- step(base_model, scope = full_scope, direction = "forward")
```

```
## Start:  AIC=3551.56
## Heating_Load ~ 1
##
##                        Df Sum of Sq   RSS    AIC
## + Overall_Height        1     61776 16314 2351.0
## + Roof_Area             1     58001 20089 2510.9
## + Surface_Area          1     33822 44267 3117.6
## + Relative_Compactness  1     30238 47852 3177.4
## + Wall_Area             1     16214 61876 3374.8
## + Glazing_Area          1      5686 72404 3495.5
## + Glazing_Distribution  1       596 77494 3547.7
## <none>                              78090 3551.6
## + Orientation           1         1 78089 3553.6
##
## Step:  AIC=2351
## Heating_Load ~ Overall_Height
##
##                        Df Sum of Sq   RSS    AIC
## + Glazing_Area          1    5686.1 10628 2023.9
## + Wall_Area             1    3589.6 12724 2162.2
## + Surface_Area          1    3275.2 13039 2180.9
## + Relative_Compactness  1    3220.7 13093 2184.1
## + Glazing_Distribution  1     596.1 15718 2324.4
## <none>                              16314 2351.0
## + Roof_Area             1      14.3 16300 2352.3
## + Orientation           1       0.5 16314 2353.0
##
## Step:  AIC=2023.88
## Heating_Load ~ Overall_Height + Glazing_Area
##
##                        Df Sum of Sq     RSS    AIC
## + Wall_Area             1    3589.6  7038.4 1709.4
## + Surface_Area          1    3275.2  7352.8 1742.9
## + Relative_Compactness  1    3220.7  7407.2 1748.6
## + Glazing_Distribution  1      73.1 10554.8 2020.6
## <none>                              10628.0 2023.9
## + Roof_Area             1      14.3 10613.6 2024.8
## + Orientation           1       0.5 10627.4 2025.8
##
## Step:  AIC=1709.38
## Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area
##
##                        Df Sum of Sq    RSS    AIC
## + Relative_Compactness  1   195.930 6842.4 1689.7
## + Roof_Area             1    79.723 6958.6 1702.6
## + Surface_Area          1    79.723 6958.6 1702.6
## + Glazing_Distribution  1    73.139 6965.2 1703.4
## <none>                              7038.4 1709.4
## + Orientation           1     0.523 7037.8 1711.3
##
## Step:  AIC=1689.7
## Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area + Relative_Compactness
```

```
##
##                         Df Sum of Sq    RSS    AIC
## + Surface_Area           1   225.004 6617.4 1666.0
## + Roof_Area              1   225.004 6617.4 1666.0
## + Glazing_Distribution  1    73.139 6769.3 1683.5
## <none>                               6842.4 1689.7
## + Orientation            1     0.523 6841.9 1691.6
##
## Step:  AIC=1666.02
## Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area + Relative_Compactness +
##     Surface_Area
##
##                         Df Sum of Sq    RSS    AIC
## + Glazing_Distribution  1    73.139 6544.3 1659.5
## <none>                               6617.4 1666.0
## + Orientation            1     0.523 6616.9 1668.0
##
## Step:  AIC=1659.49
## Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area + Relative_Compactness +
##     Surface_Area + Glazing_Distribution
##
##                Df Sum of Sq    RSS    AIC
## <none>                      6544.3 1659.5
## + Orientation  1   0.52253 6543.8 1661.4
```

```
reg_model
```

```
##
## Call:
## lm(formula = Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area +
##     Relative_Compactness + Surface_Area + Glazing_Distribution,
##     data = data)
##
## Coefficients:
##          (Intercept)        Overall_Height          Glazing_Area
##             83.93176               4.16995              19.93274
##            Wall_Area  Relative_Compactness          Surface_Area
##              0.06081             -64.77343              -0.08729
## Glazing_Distribution
##              0.20378
```

Final model

```
best_mdl <- lm(formula = Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area +
    Relative_Compactness + Surface_Area + Glazing_Distribution,
    data = data)
summary(best_mdl)
```

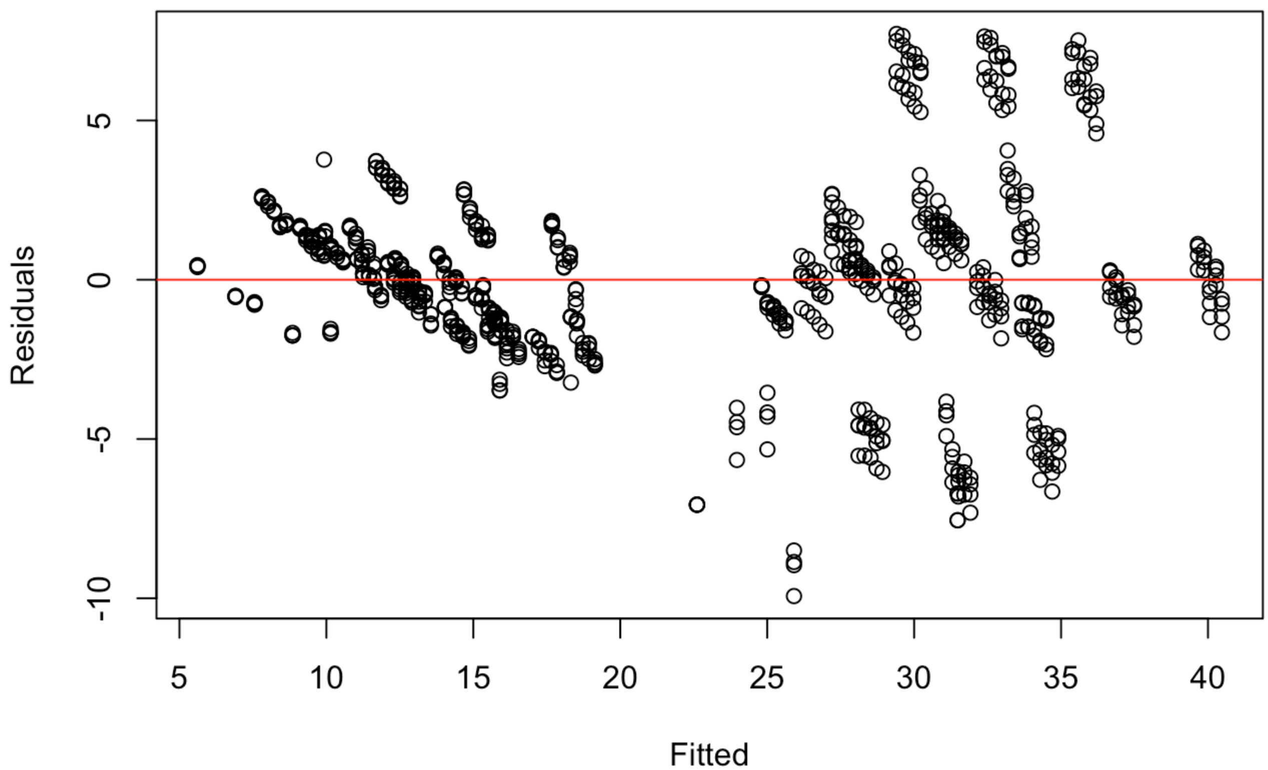
```
##
## Call:
## lm(formula = Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area +
##     Relative_Compactness + Surface_Area + Glazing_Distribution,
##     data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.9315 -1.3189 -0.0262  1.3587  7.7169
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          83.931762  19.018978   4.413 1.17e-05 ***
## Overall_Height        4.169954   0.337781  12.345  < 2e-16 ***
## Glazing_Area         19.932736   0.813484  24.503  < 2e-16 ***
## Wall_Area             0.060813   0.006644   9.153  < 2e-16 ***
## Relative_Compactness -64.773432  10.283096  -6.299 5.06e-10 ***
## Surface_Area         -0.087289   0.017065  -5.115 3.97e-07 ***
## Glazing_Distribution   0.203777   0.069875   2.916  0.00365 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.933 on 761 degrees of freedom
## Multiple R-squared:  0.9162, Adjusted R-squared:  0.9155
## F-statistic:  1387 on 6 and 761 DF,  p-value: < 2.2e-16
```

```
AIC(best_mdl)
```

```
## [1] 3840.975
```

Residual s plot

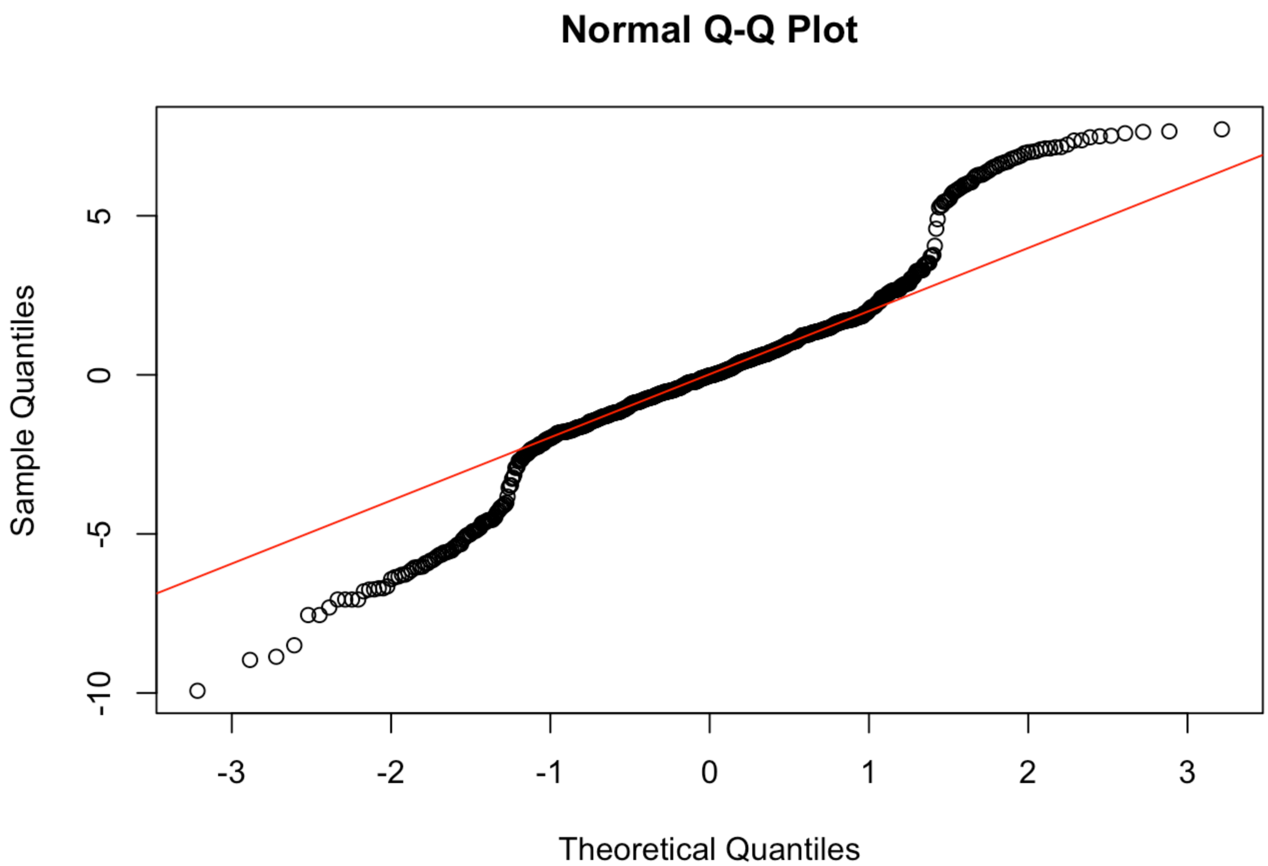
```
plot(best_mdl$fitted.values, best_mdl$residuals,
     xlab="Fitted", ylab="Residuals")
abline(h=0, col="red")
```



Q-

Q plot

```
qqnorm(best_mdl$residuals)
qqline(best_mdl$residuals, col="red")
```



##

Final checks on multicolinearity on the finl model

```
pred <- predict(best_mdl)
resid <- best_mdl$residuals

RMSE <- sqrt(mean(resid^2))
MAE <- mean(abs(resid))

RMSE
```

```
## [1] 2.919112
```

```
MAE
```

```
## [1] 2.0673
```

#Train–Test Split

```
#Splitting data
set.seed(123)
n <- nrow(data)
train_index <- sample(1:n, size = 0.75 * n)

train <- data[train_index, ]
test  <- data[-train_index, ]
```

Model fitting/training

```
train_model <- lm(
  Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area +
    Relative_Compactness + Surface_Area + Glazing_Distribution,
  data = train
)

summary(train_model)
```

```
##
## Call:
## lm(formula = Heating_Load ~ Overall_Height + Glazing_Area + Wall_Area +
##     Relative_Compactness + Surface_Area + Glazing_Distribution,
##     data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1613  -1.4466  -0.0755   1.3843   7.5966
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           86.955899  22.298511   3.900 0.000108 ***
## Overall_Height         4.338423   0.397030  10.927  < 2e-16 ***
## Glazing_Area          18.836126   0.956292  19.697  < 2e-16 ***
## Wall_Area              0.058075   0.007819   7.427 4.08e-13 ***
## Relative_Compactness -67.893684  12.079943  -5.620 2.99e-08 ***
## Surface_Area          -0.088176   0.019995  -4.410 1.24e-05 ***
## Glazing_Distribution   0.283676   0.082590   3.435 0.000636 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.985 on 569 degrees of freedom
## Multiple R-squared:  0.9152, Adjusted R-squared:  0.9143
## F-statistic:  1024 on 6 and 569 DF,  p-value: < 2.2e-16
```

#Prediction on test set

```
test$predicted <- predict(train_model, newdata = test)
```

#Evaluation of the results

```r
#Mean Squared Error (MSE)
mse <- mean((test$Heating_Load - test$predicted)^2)
mse
```

```
## [1] 7.838366
```

Root MSE

```r
rmse <- sqrt(mse)
rmse
```

```
## [1] 2.799708
```

Mean Absolute Error (MAE)

```r
mae <- mean(abs(test$Heating_Load - test$predicted))
mae
```

```
## [1] 1.967023
```

Prediction